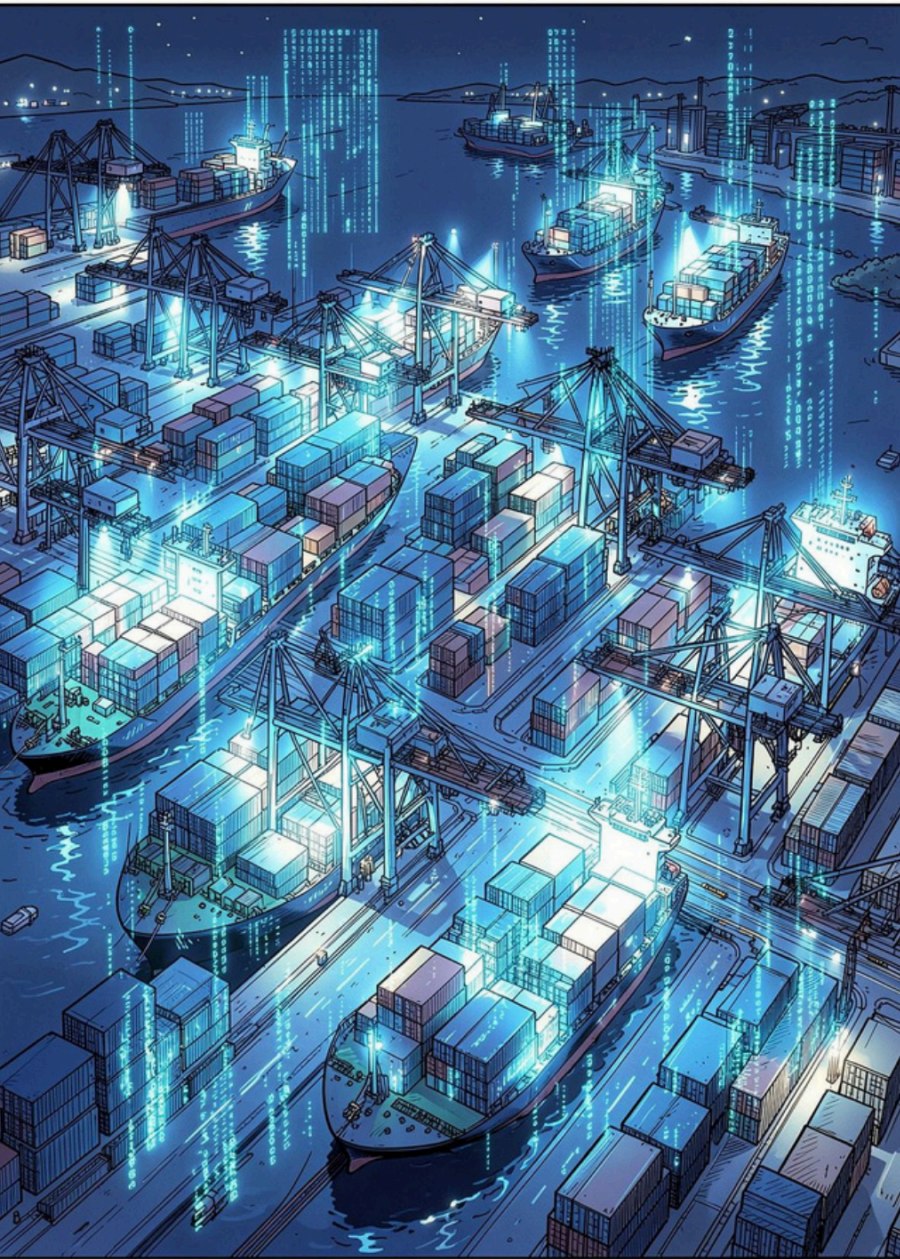




IBM BOB AI INNOVATION HACKATHON · PROBLEM L1

SMARTPORT AI

Container Congestion Predictor & Port Operations Optimiser

Predict → Explain → Optimise → Act

Team Apex

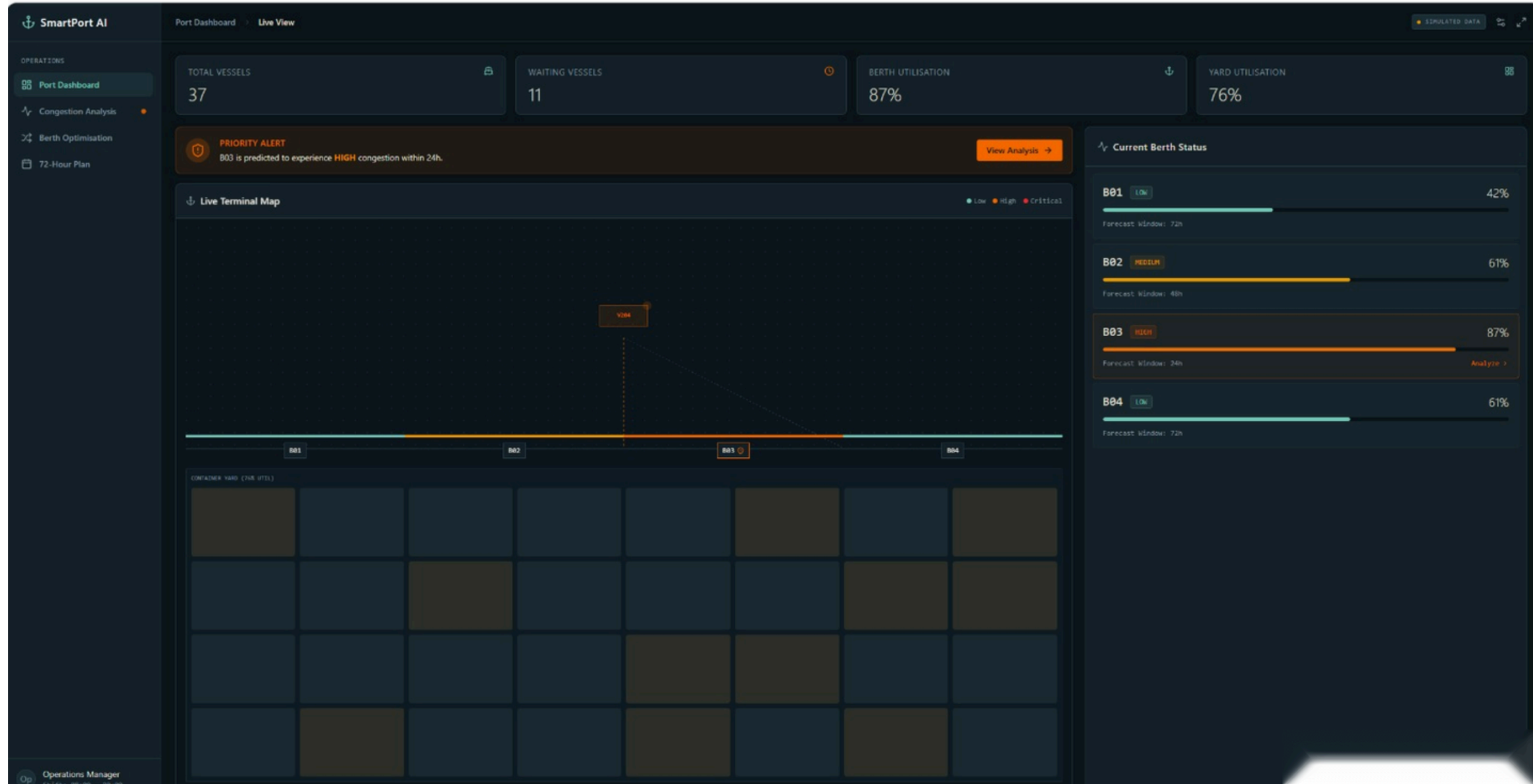
Priyam Jasoliya

Dhairya Gujarati

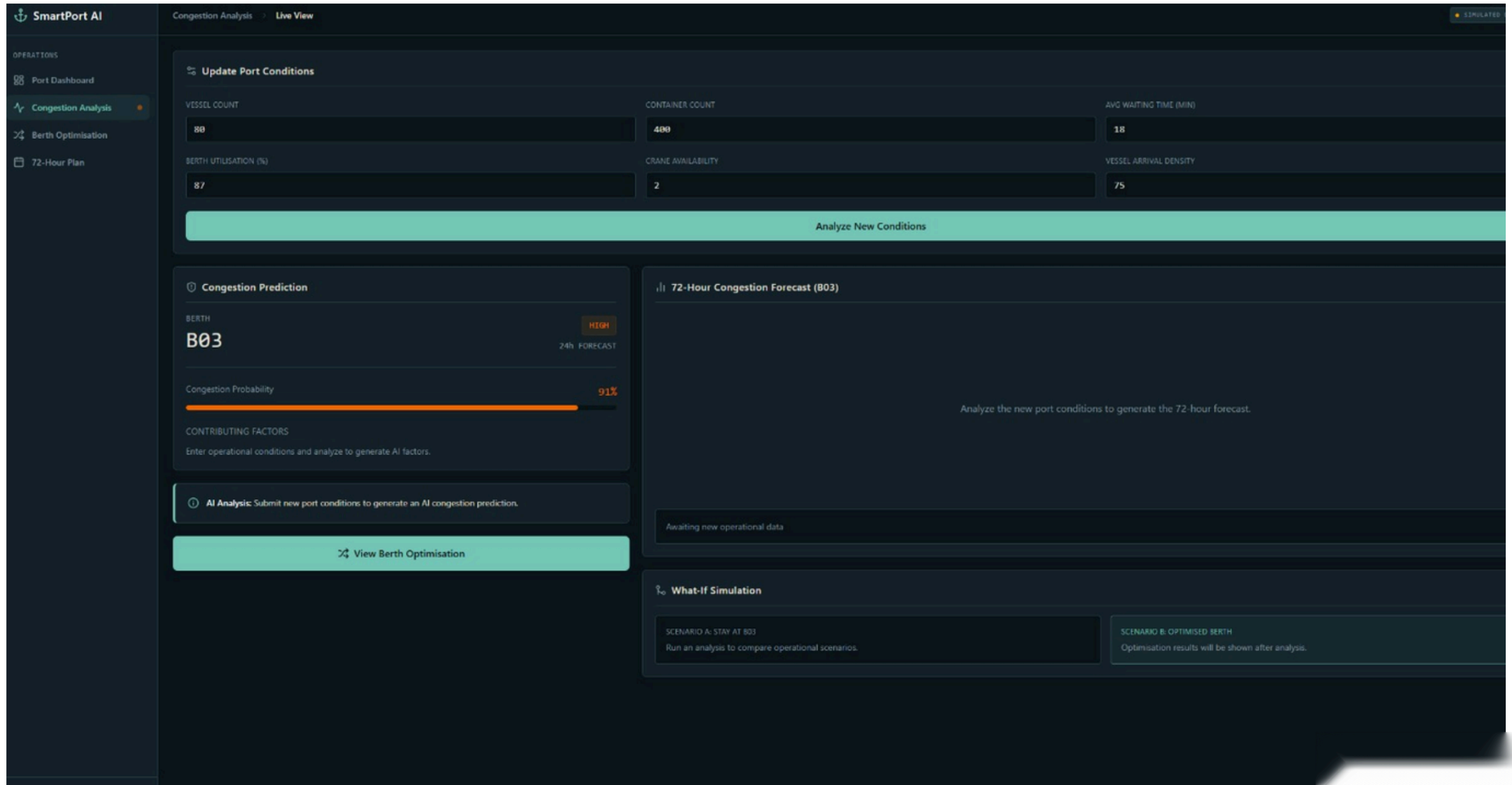
Srushti Makrubiya

Krishna Mangukiya

DASHBOARD

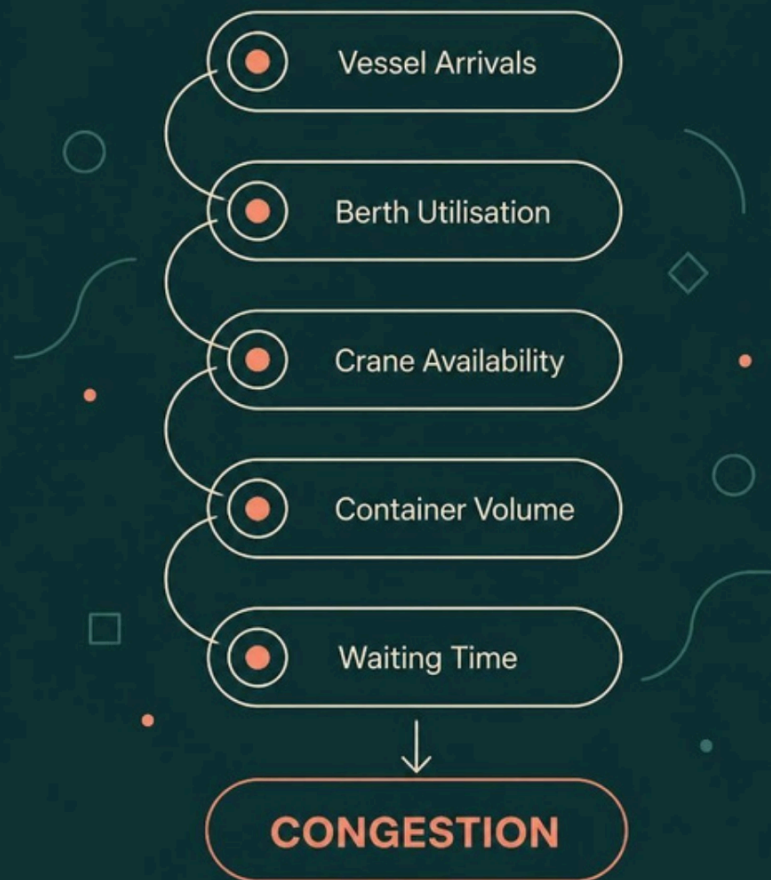


CONGESTION ANALYSIS



Port Congestion Is a Decision Problem

Every delay traces back to a missed or late decision. Ports need to see problems before they escalate.



Key Operational Challenges

- Unpredictable vessel arrival patterns
- Rising container volumes with limited berth capacity
- Crane and resource bottlenecks
- Increasing vessel waiting times
- Reactive rather than proactive operational decisions

📄 "Operators need to know what will happen — and what to do next."

OUR SOLUTION

SmartPort AI — From Prediction to Action

A single intelligent pipeline that turns raw port data into operational decisions — in four stages.



Predict

ML detects congestion risk before it escalates



Explain

Surfaces the operational factors driving the risk



Optimise

Recommends berth and resource reallocations



Act

Generates a 72-hour operational action plan

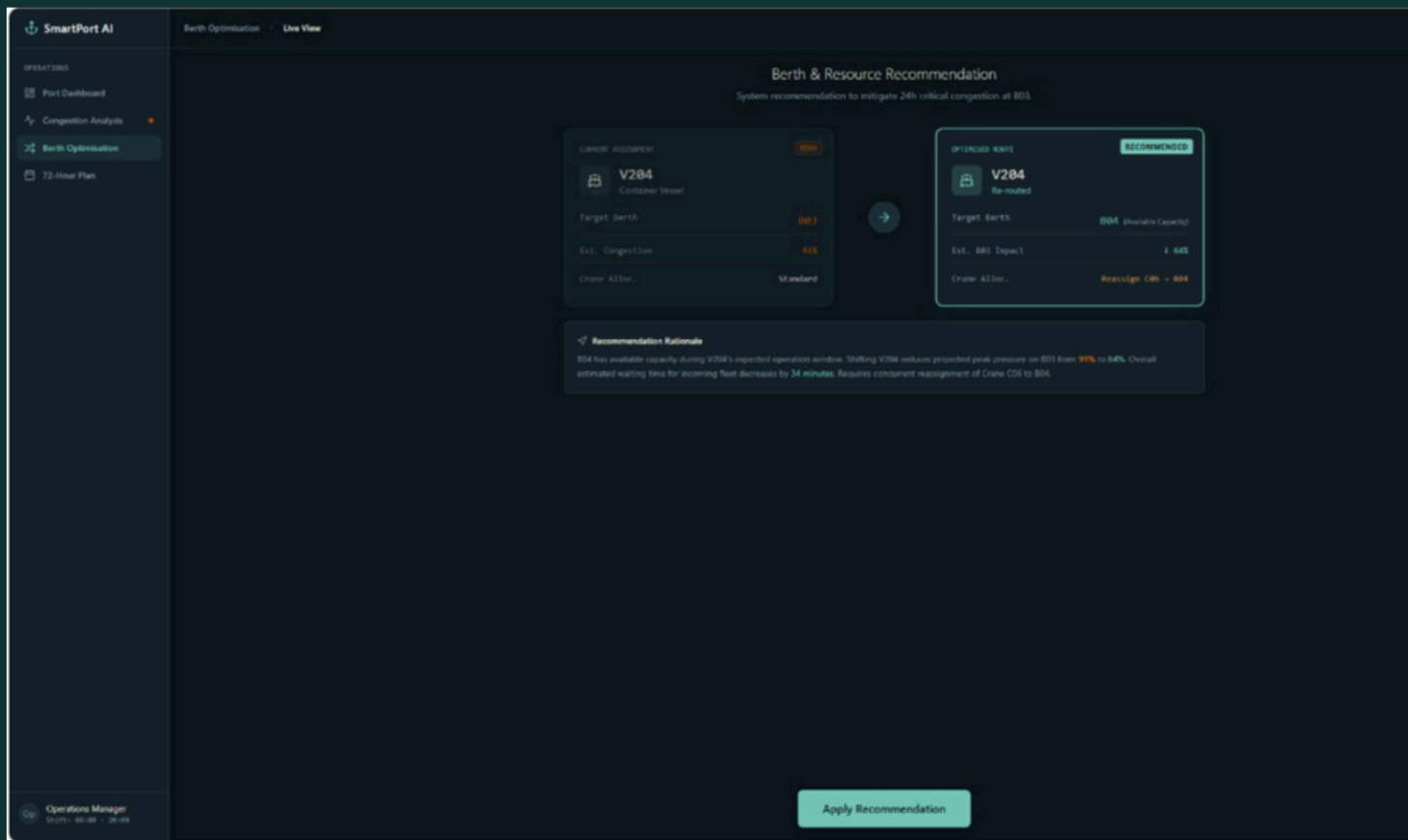
Predict — Detect Congestion Before It Escalates

Operational Inputs

- Vessel count & arrival density
- Container volume
- Berth utilisation rate
- Crane availability
- Average waiting time

ML Model

Random Forest — trained on structured operational port data to evaluate current conditions and produce a congestion-risk score.



- ❏ **86% Probability** — High Congestion Risk detected for the current operational window.

Explain – Understand What Is Driving the Risk

SmartPort AI doesn't stop at a risk score – it shows **why** congestion is occurring and how the risk may evolve over the next 72 hours.

Contributing Operational Factors

- High berth utilisation
- Elevated vessel waiting time
- Surging container volume
- Peak vessel count window
- Reduced crane availability

Prototype uses structured/simulated operational data.



Optimise — Turn Predictions Into Decisions

When congestion is predicted, SmartPort AI evaluates alternatives and recommends the best operational move.

Current Assignment

Vessel → **Berth B03**

High congestion risk · Extended wait

Recommended Action

Vessel → **Berth B04**

Lower wait · Better crane access



Berth Recommendation

Optimal berth assignment based on real-time conditions



Crane Allocation

Dynamic reassignment to reduce bottlenecks



Resource Optimisation

Maximise utilisation across available assets



Act — From Scenario Testing to an Operational Plan

What-If Simulation

Operators test alternative scenarios before committing to a decision.

01

Change operational conditions

02

Run scenario

03

Compare outcomes

04

Choose better decision

72-Hour Operational Plan

Predictions and optimisation recommendations are converted into a structured, time-bound action plan.

Prediction

Congestion risk
identified

Optimisation

Best action
recommended

Prioritised Actions

Ranked by impact

72-Hour Plan

Ready to execute

72 hour plan

SmartPort AI

OPERATIONS

Port Dashboard

Congestion Analysis

Berth Optimisation

72-Hour Plan

72-Hour Plan

Live View

SIMULATED DATA

72-Hour Operational Plan

TODAY

TIME	VESSEL	BERTH	RESOURCE	ACTION	PRIORITY
08:00	V201	B02	Crane C04	Berthing	NORMAL
10:00	V204	B03	Crane C03	Berthing	HIGH
12:30	N/A	B03	Crane C07	Maintenance window	MEDIUM
15:00	V206	B01	Crane C02	Departure	NORMAL

TOMORROW

TIME	VESSEL	BERTH	RESOURCE	ACTION	PRIORITY
09:00	V209	B01	Crane C01	Berthing	NORMAL
11:30	V211	B04	Crane C06	Berthing	MEDIUM
14:00	V213	B03	Crane C07	Departure	HIGH

DAY 3

TIME	VESSEL	BERTH	RESOURCE	ACTION	PRIORITY
09:30	V215	B02	Crane C04	Berthing	NORMAL
13:00	V219	B04	Crane C06	Berthing	HIGH
16:30	V221	B03	Crane C03	Departure	CRITICAL

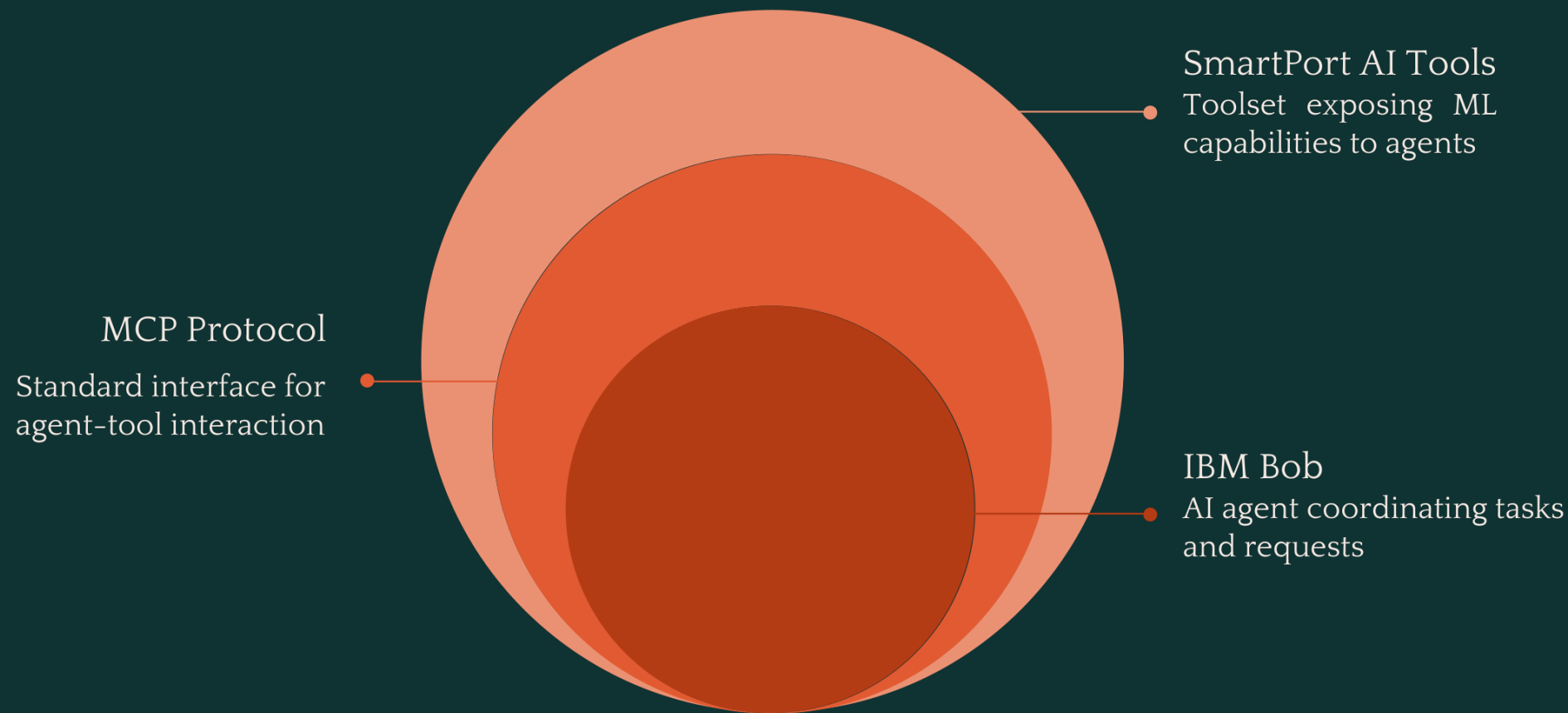
Op

Operations Manager

Shift: 00:00 - 20:00

IBM Bob + MCP – AI-Native Integration

MCP provides a standard tool interface that allows AI agents such as IBM Bob to interact with SmartPort AI capabilities.



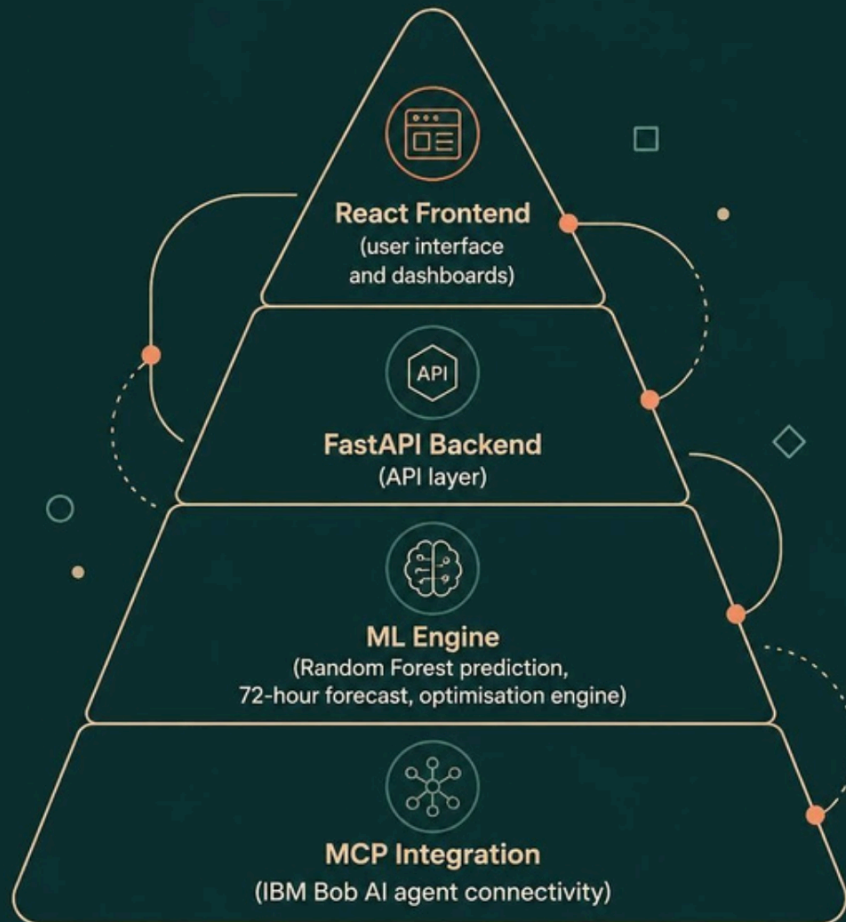
IBM Bob communicates with SmartPort AI through the MCP protocol, enabling seamless AI-agent interaction with all operational capabilities.

Capabilities Exposed via MCP

Congestion Prediction	Risk Explanation
Berth Optimisation	Resource & Crane Optimisation
What-If Simulation	72-Hour Operational Planning

Technology Behind SmartPort AI

Tech Stack



Frontend

React – responsive
operational
dashboards

Backend

Python + FastAPI –
RESTful API layer

Machine Learning

Python + Random
Forest – congestion
prediction

Optimisation

Rule and heuristic-
based operational
engine

Integration

MCP + IBM Bob – AI
agent connectivity

Data

Structured/simulated
operational data
(prototype)

From Port Prediction to Port Intelligence

Current Impact

- Earlier congestion awareness
- Explainable operational insights
- Better berth and resource decisions
- What-if scenario testing
- 72-hour operational planning
- AI-agent integration through MCP

Future Scope

- Live AIS integration
- Real port TOS connectivity
- Real-time IoT and sensor data
- Advanced optimisation algorithms
- Digital twin simulation
- Continuous learning from historical data
- Multi-port scaling

"SmartPort AI transforms port operations from reactive management into predictive, explainable, and optimised decision-making."

